SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

*I **R SARVESH / 192521143** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

R SARVESH

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C? (5 Marks)

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis. (5 Marks)

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

a. Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
b. Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.
ii) Suggest two effective techniques to reduce overfitting in this model.
iii) Discuss how overfitting affects the model's generalization ability on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
(c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
(h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%
i) Identify the epoch at which the model starts overfitting.
ii) Explain why validation accuracy decreases despite training accuracy increasing.
iii) Suggest two methods to improve validation performance.

6. A machine learning model shows the following loss values during training:
(5 Marks)

(a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
(b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3
(c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
(d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
(e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
(f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5
(g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8
(h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1
i) At which epoch does the model achieve optimal generalization?
ii) Identify signs of overfitting from the data.
iii) Recommend two regularization techniques and explain their impact.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires). (5 Marks)

a. Front identified as Front: 119
b. Front identifies as Back: 06
c. Back identified as Back: 120
d. Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

a. Diseased identified as Diseased: 138
b. Diseased identified as Healthy: 12
c. Healthy identified as Healthy: 135
d. Healthy identified as Diseased: 15

Compute: **Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.**

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective): (5 Marks)

- Defective identified as Defective: 2 30
- Defective identified as Non-Defective: 20
- Non-Defective identified as Non-Defective: 225
- Non-Defective identified as Defective: 25

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated. (5 Marks)

12. A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective. (5 Marks)

13. In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150.The dataset contains 1000 samples per class, and 20% of the data is used for testing. Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity. (5 Marks)

14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data.
Calculate: Accuracy Precision Sensitivity Specificity. (5 Marks)

15. A deep learning model is trained for a classification task, and the following loss values are observed: (5 Marks)

i. Epoch 1: Training Loss = 2.8, Validation Loss = 2.6
ii. Epoch 2: Training Loss = 2.4, Validation Loss = 2.2
iii. Epoch 3: Training Loss = 2.1, Validation Loss = 1.9
iv. Epoch 4: Training Loss = 1.9, Validation Loss = 1.8
v. Epoch 5: Training Loss = 1.7, Validation Loss = 1.9
vi. Epoch 6: Training Loss = 1.5, Validation Loss = 2.1

vii. Epoch 7: Training Loss = 1.3, Validation Loss = 2.4
viii. Epoch 8: Training Loss = 1.2, Validation Loss = 2.7

Answer the following:

a. At which epoch does the model achieve the best generalization performance?
b. At which point does overfitting begin? Justify your answer.
c. Suggest two techniques to reduce overfitting and briefly explain how they help.

Q1. Bayes' Theorem – Probability of Selecting Box C

Given:

Box	Red	Blue	Green	Total
A	6	4	2	12
B	5	7	3	15
C	8	2	5	15

Box B is twice as likely to be selected as A and C.

Therefore,

P(A) = 1/4

P(B) = 2/4 = 1/2

P(C) = 1/4

Probability of drawing a blue ball:

P(Blue|A) = 4/12 = 1/3

P(Blue|B) = 7/15

P(Blue|C) = 2/15

Using Bayes' theorem:

P(C|Blue) = [P(C)P(Blue|C)]/[P(A)P(Blue|A) + P(B)P(Blue|B) + P(C)P(Blue|C)]

= [(1/4)(2/15)]/[(1/4)(1/3) + (1/2)(7/15) + (1/4)(2/15)]

= (2/60)/(1/12 + 7/30 + 1/30)

= (1/30)/(5/60 + 14/60 + 2/60)

= (1/30)/(21/60)

= (1/30)(60/21)

= 2/21

Therefore,

P(C|Blue) = 2/21 = 0.0952 = 9.52%

Final Answer:

The probability that the selected box was Box C is 2/21 or 9.52%.

Q2. Bayes' Theorem – Probability of Section C

Given:

Section	Red	Blue	Black	Total
A	5	3	2	10
B	4	6	5	15
C	7	2	6	15

Section B is twice as likely as A and C.

Therefore,

P(A) = 1/4

P(B) = 1/2

P(C) = 1/4

Conditional probabilities:

P(Blue|A) = 3/10

P(Blue|B) = 6/15 = 2/5

P(Blue|C) = 2/15

Using Bayes' theorem:

P(C|Blue) = [P(C)P(Blue|C)]/[P(A)P(Blue|A) + P(B)P(Blue|B) + P(C)P(Blue|C)]

= [(1/4)(2/15)]/[(1/4)(3/10) + (1/2)(2/5) + (1/4)(2/15)]

= (1/30)/(3/40 + 1/5 + 1/30)

Taking LCM = 120:

= (1/30)/(9/120 + 24/120 + 4/120)

= (1/30)/(37/120)

= (1/30)(120/37)

= 4/37

Therefore,

P(C|Blue) = 4/37 = 0.1081 = 10.81%

Final Answer:

The probability that the car came from Section C is 4/37 or 10.81%.

Q3. Confusion Matrix – Medical Image Classification

Given:

There are 1800 samples per class.

Total samples:

Total = 1800 + 1800 = 3600

20% is reserved for testing.

Test samples:

Test = 20% of 3600

= (20/100)(3600)

= 720

Therefore, the testing dataset contains **720 samples**.

Given:

TP = 210

TN = 190

FP = 30

We need to find FN.

Total test samples:

TP + TN + FP + FN = 720

Therefore,

FN = 720 - TP - TN - FP

= 720 - 210 - 190 - 30

= 290

Confusion Matrix

Actual / Predicted	**Positive**	**Negative**
Positive	TP = 210	FN = 290
Negative	FP = 30	TN = 190

1. Accuracy

Formula:

Accuracy = (TP + TN)/(TP + TN + FP + FN)

= (210 + 190)/(210 + 190 + 30 + 290)

= 400/720

= 0.5556

Accuracy = 55.56%

2. Precision

Formula:

Precision = TP/(TP + FP)

= 210/(210 + 30)

= 210/240

= 0.875

Precision = 87.50%

This means that among the samples predicted as malignant, 87.5% were actually malignant.

3. Sensitivity / Recall

Formula:

Sensitivity = TP/(TP + FN)

= 210/(210 + 290)

= 210/500

= 0.42

Sensitivity = 42%

Sensitivity measures the ability of the model to correctly identify malignant cases.

4. Specificity

Formula:

Specificity = TN/(TN + FP)

= 190/(190 + 30)

= 190/220

= 0.8636

Specificity = 86.36%

Interpretation

The model has **high precision and specificity**, meaning that its positive predictions are relatively reliable and it correctly identifies many benign cases.

However, its **sensitivity is only 42%**, meaning that many malignant cases are missed. In medical diagnosis, this is important because False Negatives can be dangerous.

Therefore, based on the given test results, the model requires improvement, particularly in **sensitivity/recall**, before it can be considered reliable for detecting breast cancer.

Final Answers:

Metric	Value
FN	290
Accuracy	55.56%
Precision	87.50%
Sensitivity	42%
Specificity	86.36%

Q4. Training Loss and Validation Loss – Overfitting

Given:

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

i) Epoch at which overfitting begins

Training loss continuously decreases:

$2.8 \rightarrow 2.5 \rightarrow 2.2 \rightarrow 2.0 \rightarrow 1.8 \rightarrow 1.6 \rightarrow 1.5 \rightarrow 1.3$

Validation loss decreases initially:

$2.4 \rightarrow 2.2 \rightarrow 2.0$

The minimum validation loss occurs at **Epoch 3**.

After Epoch 3, validation loss begins to increase while training loss continues to decrease.

Therefore:

Overfitting begins after Epoch 3, i.e., from Epoch 4.

ii) Two techniques to reduce overfitting

1. Dropout

Dropout randomly disables some neurons during training. This prevents the neural network from depending too heavily on particular neurons and improves generalization.

2. Early Stopping

Training is stopped when validation loss starts increasing. In this case, training could be stopped around Epoch 3 because validation loss is minimum there.

iii) Effect of overfitting on generalization

Overfitting occurs when the model learns the training data too specifically, including noise and unnecessary patterns.

In this example, training loss continues to decrease, but validation loss increases after Epoch 3.

Therefore, the model performs well on training data but performs poorly on unseen data.

Hence, overfitting reduces the model's generalization ability.

Final Answer:

Overfitting starts from Epoch 4.

Two methods:

1. Dropout
2. Early Stopping

Q5. Training Accuracy and Validation Accuracy – Overfitting

Given:

Epoch	Training Accuracy	Validation Accuracy
1	60%	58%
2	65%	63%
3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

i) Epoch at which overfitting starts

Validation accuracy increases until Epoch 5:

58% → 63% → 68% → 72% → **73%**

After Epoch 5, validation accuracy decreases:

73% → 72% → 70% → 68%

At the same time, training accuracy continues to increase.

Therefore:

Overfitting starts after Epoch 5, i.e., from Epoch 6.

ii) Why does validation accuracy decrease?

The training accuracy increases because the model continues learning the training data.

However, after Epoch 5, the model starts learning patterns that are specific to the training data, including noise.

Therefore:

- Training accuracy increases.
- Validation accuracy decreases.
- The gap between training and validation accuracy increases.

This indicates **overfitting**.

iii) Two methods to improve validation performance

1. Dropout

Dropout randomly removes neurons during training. It prevents excessive dependence on particular features and improves generalization.

2. Early Stopping

Training should be stopped when validation accuracy stops improving or validation loss starts increasing.

In this case, training can be stopped around **Epoch 5**.

Final Answer:

Overfitting begins from Epoch 6.

Two methods:

1. Dropout
2. Early stopping

Q6. Training Loss and Validation Loss – Optimal Generalization

Given:

Epoch	Training Loss	Validation Loss
1	3.0	2.7
2	2.6	2.3
3	2.3	2.1
4	2.1	2.0
5	1.9	2.2
6	1.7	2.5
7	1.6	2.8
8	1.4	3.1

i) Epoch of optimal generalization

For good generalization, validation loss should be minimum.

Validation losses are:

2.7, 2.3, 2.1, **2.0**, 2.2, 2.5, 2.8, 3.1

The minimum validation loss is **2.0 at Epoch 4**.

Therefore:

Optimal generalization occurs at Epoch 4.

ii) Signs of overfitting

At Epoch 4:

Training Loss = 2.1

Validation Loss = 2.0

After Epoch 4:

Training loss continues to decrease:

2.1 → 1.9 → 1.7 → 1.6 → 1.4

But validation loss increases:

2.0 → 2.2 → 2.5 → 2.8 → 3.1

This difference between training and validation performance is a clear indication of **overfitting**.

Therefore:

Overfitting starts after Epoch 4, i.e., from Epoch 5.

iii) Two regularization techniques

1. L2 Regularization

L2 regularization adds a penalty for large model weights to the loss function.

This prevents the model from becoming unnecessarily complex and improves generalization.

2. Dropout

Dropout randomly disables neurons during training.

This prevents the model from depending too much on particular neurons and reduces overfitting.

Final Answer:

Optimal generalization = Epoch 4

Overfitting begins = Epoch 5

Regularization techniques:

1. L2 Regularization
2. Dropout

Q7. AI Robot – Front and Back Tire Classification

Given:

Total tires = 250

Actual Front = 125

Actual Back = 125

Actual / Predicted	**Front**	**Back**
Front	119	6
Back	5	120

Considering **Front** as the positive class:

TP = 119

FN = 6

TN = 120

FP = 5

1. Accuracy

Formula:

Accuracy = (TP + TN)/(TP + TN + FP + FN)

= (119 + 120)/(119 + 120 + 5 + 6)

= 239/250

= 0.956

Accuracy = 95.6%

2. Precision

Formula:

Precision = TP/(TP + FP)

= 119/(119 + 5)

= 119/124

= 0.9597

Precision = 95.97%

3. Sensitivity / Recall

Formula:

Sensitivity = TP/(TP + FN)

= 119/(119 + 6)

= 119/125

= 0.952

Sensitivity = 95.2%

4. Specificity

Formula:

Specificity = TN/(TN + FP)

= 120/(120 + 5)

= 120/125

= 0.96

Specificity = 96%

5. F1-Score

Formula:

F1-Score = 2(Precision × Recall)/(Precision + Recall)

= 2[(119/124)(119/125)]/[(119/124) + (119/125)]

= 0.9558

F1-Score ≈ 95.58%

Final Answer:

Metric	Value
Accuracy	95.6%
Precision	95.97%
Sensitivity	95.2%
Specificity	96%
F1-Score	95.58%

Q8. AI Medical Diagnosis – Disease Classification

Considering **Diseased** as the positive class.

Given:

Actual / Predicted	Diseased	Healthy
Diseased	138	12
Healthy	15	135

Therefore:

TP = 138

FN = 12

TN = 135

FP = 15

Total = 300

1. Accuracy

Formula:

Accuracy = (TP + TN)/(TP + TN + FP + FN)

= (138 + 135)/300

= 273/300

= 0.91

Accuracy = 91%

2. Precision

Formula:

Precision = TP/(TP + FP)

= 138/(138 + 15)

= 138/153

= 0.9020

Precision = 90.20%

3. Sensitivity / Recall

Formula:

Sensitivity = TP/(TP + FN)

= 138/(138 + 12)

= 138/150

= 0.92

Sensitivity = 92%

4. Specificity

Formula:

Specificity = TN/(TN + FP)

= 135/(135 + 15)

= 135/150

= 0.90

Specificity = 90%

5. F1-Score

Formula:

F1-Score = 2(Precision × Recall)/(Precision + Recall)

= 2[(138/153)(138/150)]/[(138/153) + (138/150)]

= 0.9109

F1-Score ≈ 91.09%

Final Answer:

Metric	Value
Accuracy	91%
Precision	90.20%
Sensitivity	92%
Specificity	90%
F1-Score	91.09%

Q9. Spam Email Classification

Considering **Spam** as the positive class.

Given:

Actual / Predicted	Spam	Not Spam
Spam	180	20
Not Spam	30	170

Therefore:

TP = 180

FN = 20

TN = 170

FP = 30

Total = 400

1. Accuracy

Formula:

Accuracy = (TP + TN)/(TP + TN + FP + FN)

= (180 + 170)/400

= 350/400

= 0.875

Accuracy = 87.5%

2. Precision

Formula:

Precision = TP/(TP + FP)

= 180/(180 + 30)

= 180/210

= 0.8571

Precision = 85.71%

3. Sensitivity / Recall

Formula:

Sensitivity = TP/(TP + FN)

= 180/(180 + 20)

= 180/200

= 0.90

Sensitivity = 90%

4. Specificity

Formula:

Specificity = TN/(TN + FP)

= 170/(170 + 30)

= 170/200

= 0.85

Specificity = 85%

5. F1-Score

Formula:

F1-Score = 2(Precision × Recall)/(Precision + Recall)

= 2[(180/210)(180/200)]/[(180/210) + (180/200)]

= 0.8780

F1-Score ≈ 87.80%

Final Answer:

Metric	Value
Accuracy	87.5%
Precision	85.71%

Sensitivity	90%
Specificity	85%
F1-Score	87.80%

Q10. Manufacturing Quality Inspection

Considering **Defective** as the positive class:

Given:

Actual / Predicted	**Defective**	**Non-Defective**
Defective	230	20
Non-Defective	25	225

Therefore:

TP = 230

FN = 20

TN = 225

FP = 25

Total = 500

1. Accuracy

Formula:

Accuracy = (TP + TN)/(TP + TN + FP + FN)

= (230 + 225)/500

= 455/500

= 0.91

Accuracy = 91%

2. Precision

Formula:

Precision = TP/(TP + FP)

= 230/(230 + 25)

= 230/255

= 0.9020

Precision = 90.20%

3. Sensitivity / Recall

Formula:

Sensitivity = TP/(TP + FN)

= 230/(230 + 20)

= 230/250

= 0.92

Sensitivity = 92%

4. Specificity

Formula:

Specificity = TN/(TN + FP)

= 225/(225 + 25)

= 225/250

= 0.90

Specificity = 90%

5. F1-Score

Formula:

F1-Score = 2(Precision × Recall)/(Precision + Recall)

= 2[(230/255)(230/250)]/[(230/255) + (230/250)]

= 0.9109

F1-Score ≈ 91.09%

Final Answer:

Metric	Value
Accuracy	91%
Precision	90.20%
Sensitivity	92%
Specificity	90%
F1-Score	91.09%

Q11. Bayes' Theorem – Shop X and Shop Y

Given:

Shop	Pure Oil	Adulterated Oil	Total
X	20	30	50
Y	40	20	60

Shop Y is twice as likely to be selected as Shop X.

Therefore,

$P(X) = 1/3$

$P(Y) = 2/3$

The selected bottle is found to be **adulterated**.

Therefore,

$P(\text{Adulterated}|X) = 30/50 = 3/5$

$P(\text{Adulterated}|Y) = 20/60 = 1/3$

Probability that the bottle came from Shop X:

Formula:

$P(X|\text{Adulterated}) = [P(X)P(\text{Adulterated}|X)]/[P(X)P(\text{Adulterated}|X) + P(Y)P(\text{Adulterated}|Y)]$

Substituting:

$= [(1/3)(3/5)]/[(1/3)(3/5) + (2/3)(1/3)]$

$= (1/5)/(1/5 + 2/9)$

$= (1/5)/(9/45 + 10/45)$

$= (1/5)/(19/45)$

$= (1/5)(45/19)$

$= 9/19$

Therefore,

$P(X|\text{Adulterated}) = 9/19 = 47.37\%$

Probability that the bottle came from Shop Y:

$P(Y|\text{Adulterated}) = [P(Y)P(\text{Adulterated}|Y)]/[P(X)P(\text{Adulterated}|X) + P(Y)P(\text{Adulterated}|Y)]$

$= [(2/3)(1/3)]/[(1/3)(3/5) + (2/3)(1/3)]$

$= (2/9)/(1/5 + 2/9)$

$= (2/9)/(19/45)$

$= (2/9)(45/19)$

$= 10/19$

Therefore,

P(Y|Adulterated) = 10/19 = 52.63%

Final Answer:

Probability that the adulterated bottle came from Shop X = 9/19 = 47.37%

Probability that the adulterated bottle came from Shop Y = 10/19 = 52.63%

Q12. Bayes' Theorem – Defective Item

Given:

Section	Defective	Non-Defective	Total
A	15	35	50
B	25	25	50

Section A is **three times as likely** to be selected as Section B.

Therefore,

P(A) = 3/4

P(B) = 1/4

Conditional probabilities:

P(Defective|A) = 15/50 = 3/10

P(Defective|B) = 25/50 = 1/2

Probability that the item came from Section A

Using Bayes' theorem:

P(A|Defective) = [P(A)P(Defective|A)]/[P(A)P(Defective|A) + P(B)P(Defective|B)]

= [(3/4)(3/10)]/[(3/4)(3/10) + (1/4)(1/2)]

= (9/40)/(9/40 + 1/8)

= (9/40)/(9/40 + 5/40)

= (9/40)/(14/40)

= 9/14

Therefore,

P(A|Defective) = 9/14 = 64.29%

Probability that the item came from Section B

P(B|Defective) = [P(B)P(Defective|B)]/[P(A)P(Defective|A) + P(B)P(Defective|B)]

= (1/8)/(9/40 + 1/8)

= (5/40)/(14/40)

= 5/14

Therefore,

P(B|Defective) = 5/14 = 35.71%

Final Answer:

Probability that the defective item came from Section A = 9/14 = 64.29%

Probability that the defective item came from Section B = 5/14 = 35.71%

Q13. Binary Classification – Deep Learning + KNN

Given:

Each class contains = 1000 samples

Total dataset:

Total = 1000 + 1000 = 2000

20% is used for testing.

Test data:

Test = 20% of 2000

= (20/100)(2000)

= 400

Given:

TP = 120

TN = 150

Since the test dataset contains 400 samples:

TP + TN + FP + FN = 400

Therefore,

FP + FN = 400 - 120 - 150

= 130

Accuracy = (TP + TN)/(Total Test Samples)

= (120 + 150)/400

= 270/400

= 0.675

Accuracy = 67.5%

However, for the remaining measures:

Precision = TP/(TP + FP)

= 120/(120 + FP)

Sensitivity = TP/(TP + FN)

= 120/(120 + FN)

Specificity = TN/(TN + FP)

= 150/(150 + FP)

Final Answer:

Metric	Result
Test samples	400
TP	120
TN	150
FP + FN	130
Accuracy	67.5%

Q14. Medical Image Classification – Deep Learning + KNN

Given:

Each class contains = 1200 samples

Total samples:

Total = 1200 + 1200

= 2400

25% is used for testing.

Therefore:

Test samples = (25/100)(2400)

= 600

Given:

TP = 180

TN = 160

Therefore:

TP + TN + FP + FN = 600

180 + 160 + FP + FN = 600

FP + FN = 260

1. Accuracy

Formula:

Accuracy = (TP + TN)/(TP + TN + FP + FN)

= (180 + 160)/600

= 340/600

= 17/30

= 0.5667

Accuracy = 56.67%

2. Precision

Formula:

Precision = TP/(TP + FP)

= 180/(180 + FP)

Cannot be uniquely determined because FP is not given.

3. Sensitivity

Formula:

Sensitivity = TP/(TP + FN)

= 180/(180 + FN)

Cannot be uniquely determined because FN is not given.

4. Specificity

Formula:

Specificity = TN/(TN + FP)

= 160/(160 + FP)

Cannot be uniquely determined because FP is not given.

Final Answer:

Metric	Value
Test samples	600
TP	180
TN	160
FP + FN	260
Accuracy	56.67%

Q15. Deep Learning – Training and Validation Loss

Given:

Epoch	Training Loss	Validation Loss
1	2.8	2.6
2	2.4	2.2
3	2.1	1.9
4	1.9	1.8
5	1.7	1.9
6	1.5	2.1
7	1.3	2.4
8	1.2	2.7

a. Epoch at which the model achieves the best generalization

In Machine Learning, **lower validation loss indicates better performance on unseen/validation data**.

Validation losses are:

2.6 → 2.2 → 1.9 → **1.8** → 1.9 → 2.1 → 2.4 → 2.7

The minimum validation loss is **1.8 at Epoch 4**.

Therefore:

Best generalization performance occurs at Epoch 4.

b. Point at which overfitting begins

Up to Epoch 4, both training loss and validation loss decrease.

At Epoch 4:

Training Loss = 1.9

Validation Loss = 1.8

After Epoch 4:

Training loss continues to decrease:

1.9 → 1.7 → 1.5 → 1.3 → 1.2

But validation loss increases:

1.8 → 1.9 → 2.1 → 2.4 → 2.7

This indicates that the model is learning the training data increasingly well but its performance on unseen data is getting worse.

Therefore:

Overfitting begins after Epoch 4, i.e., from Epoch 5.

Justification:

The key sign of overfitting is:

Training Loss ↓ while Validation Loss ↑

This pattern starts after Epoch 4.

c. Two techniques to reduce overfitting

1. Dropout

Dropout randomly deactivates some neurons during training.

It prevents the neural network from becoming too dependent on particular neurons and helps the model learn more general patterns.

2. Early Stopping

Early stopping terminates training when validation performance stops improving.

For this model, the minimum validation loss occurs at Epoch 4. Therefore, training can be stopped around Epoch 4 rather than continuing until Epoch 8.

Final Answer:

Question	Answer
Best generalization	Epoch 4
Overfitting begins	Epoch 5
Reason	Training loss decreases while validation loss increases
Technique 1	Dropout

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand